Lalithsai Posam

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EDUCATION

University of California, Berkeley **Expected Graduation May 2027**

Bachelor of Science in Electrical Engineering and Computer Science (EECS), GPA: 3.80

Coursework: Data Structures/Algorithms, Databases, Computer Security, Efficient Algorithms, Discrete Mathematics

PROFESSIONAL EXPERIENCE

OpenAI **May 2026 - August 2026**

Member of Technical Staff Intern

- Built triggers infrastructure enabling Codex agent threads to wake and act on external Slack, GitHub, Outlook, and Microsoft Teams events, processing 100K+ events/day across many internal enterprise workspaces
- Refactored skills infrastructure by removing a legacy auto-switcher that consumed GPU capacity, cutting per-request GPU usage by 31%, and implemented a chat-to-work handoff instead serving 74% of enterprise workspace users
- Developed a Google Drive security control that fetches file permission metadata once and validates it against an allowlist, replacing 275K+ daily internal sharing-permission checks in Codex and reducing Codex Drive tool-call latency by 60%
- Created a Zendesk plugin for Codex to triage and resolve support tickets, launching to 10+ alpha enterprise accounts

Sparta **December 2025 - Present**

Co-Founder & CTO, a16z Speedrun (0.4% acceptance rate)

- Co-founded Sparta and built Worlds, an agent test environment running stateful twins of 97 production SaaS APIs that grades agents on database state instead of transcripts, cutting false-pass rate 45% and lifting agent task accuracy 30%
- Cut agent debug time 70% with byte-exact run replay, 8 injectable failure modes, and a 45-day virtual billing clock
- Built world initialization that boots a stateful API twin in under 50 ms from bundled sample or pseudonymized production data, identical on every run so evals never inherit state from prior executions, driving quicker agent validation
- Designed the mark-and-diff engine catching dollar-level billing errors across Stripe, Shopify, Zendesk and Salesforce twins

Pinterest, Monetization Team **May 2025 - August 2025**

Software Engineering Intern

- Implemented a real-time monitoring and alerting system for self-serve product feed-to-API migrations, used by 100+ engineers to proactively detect product catalog failures across 25+ merchants, including Walmart, Amazon, and Target
- Developed diagnostic tooling with 60+ alert rules and 10+ dashboard components to track ingestion completion, retry rates, and failure causes, reducing debugging latency and improving API transparency for merchant partners
- Integrated migration tracking into Pinterest infra (Pacer, Spinner, KvStore), supporting 500K+ product items/week

Cisco, ML Applications for Distributed Systems **May 2024 - August 2024**

Software Engineering Intern

- Developed an LLM-powered RAG chatbot using LangChain and GPT-4 to index over 5K documents and assist 150+ engineers in debugging Cisco Spitfire issues by retrieving relevant internal documentation for developers
- Engineered a knowledge graph using a GPT graph transformer to convert 3000+ network logs and 100+ SDK/XR objects into structured event relationships, reducing bug resolution time by 40% for network engineers
- Built a full-stack tool using Flask and JavaScript to visualize over 1250 graph-based SDK/XR relationships and automate test case execution, identifying 500+ unique vs. recurring bugs to speed up bug triage processes

PERSONAL PROJECTS

High-Throughput Streaming Engine

- Built C++/Python ingest pipeline with lock-free ring buffers sustaining 100K events/sec at less than 200 microseconds
- Implemented in-memory C++ limit order book with price-time priority matching, cancel/partial-fill support, and real-time trade-event logging to Prometheus, achieving less than 1 ms end-to-end latency under 50K active orders
- Added Python streaming ML-alpha overlay using River logistic regression for improved real-time event predictions

ADDITIONAL

Programming Languages: Python, Java, C++, Rust, MATLAB, JavaScript, HTML/CSS, SQL

Frameworks/Libraries: NumPy, SciPy, Pandas, TensorFlow, PyTorch, Keras, Node, React, Flask, Langchain, Kafka, River

Tools/Platforms: Git, Docker, Jupyter, Postman, Linux, Jenkins, LanceDB, BigQuery, Spark, Ollama, Prometheus